# DATA SHEET

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Subject to change Rev February 2017

# Phantom® Miro® C110

The Cost Effective and Easy to Use camera, perfect for many applications


Miro C110
Side view

*Miro C110 Rear view of connectors*

## Key Benefits:

**When it's too fast to see, and too important not to®**

The **Phantom Miro C110** is the **perfect go-to camera** for many applications that benefit from high quality, high speed analysis. Its ideal mix of **speed, image quality and ease of use** makes it one of the most flexible analytical tools in the lab.

- 915 frames per second (fps) at full 1.3 Mpx resolution, 1295 fps @ 1280 x 720, and up to 52,445 fps at smaller resolutions
- Sturdy and small, easy to connect and use, with common BNC and Ethernet connectors
- Phantom quality image and features, in a cost effective camera

The smart companion for a **wide range of applications** such as industrial trouble-shooting, mechanical analysis and motion analysis, the Miro C110 is **easy to work with**. Its sturdy, small body connects to signaling with common BNC cables, and uses a standard Ethernet cable for camera control and data downloading. Its accessible Trigger BNC and two AUX BNC's make it easily controlled in any situation.

## Specifications

The Miro C110 has a 1.3 Mpx CMOS sensor with 5.6 µm pixels and 12-bit pixel depth. It has 1.2 Gpx throughput, providing 915 fps at full resolution of 1280 x 1024, with very low noise to capture critical details. To help **eliminate motion blur**, the minimum exposure time is 5 µs. The minimum frame rate at all resolutions is 100 fps.

The image sensor format is 2/3", and takes advantage of a large selection of C and CS lenses. Available in either Color or Monochrome, **it makes the most out of available light**, with light sensitivity ISO ratings measured according to ISO 122326:2006 method:

| | D (Daylight) | T (Tungsten) |
|---|---|---|
| Monochrome | 2500 | 5000 |
| Color | 640 | 640 |

## Key Features:

12-bit 1.3 Megapixel CMOS sensor

915 fps @ 1280 x 1024
1295 fps at 1280 x 720

Available in Monochrome or Color

ISO Mono 5,000 (T), 2,500 (D)*
Color 640 (T)*, 640 (D)*, adjustable

Compact and sturdy

Standard BNC and Ethernet connections

Reversible mount for C & CS lenses

8GB of RAM included

Easy-to-use signals:
- Trigger
- AUX 1:
  - FSYNC
  - Strobe
  - Event
  - Memgate
- AUX 2:
  - Strobe
  - Ready

** Measured according to ISO 122326:2006 method*

AMETEK®
MATERIALS ANALYSIS DIVISION

# DATA SHEET

## Phantom® Miro® C110

### Complete with Easy-to-Use Phantom Features


*Phantom Miro C110*

The Miro C110 has 8GB of memory, providing almost 4.5 seconds of record time at maximum frame rate and resolution, and longer at reduced rates and resolutions.

It uses powerful Phantom Camera Control (PCC) software. PCC makes an easy job out of capturing the event and then adjusting image characteristics like color and brightness. Flexible triggering options **easily capture the images of a specific event**. Images are automatically recorded into a circular buffer, and the trigger option determines which of those images are saved to RAM – images before, after, or on either side of the trigger, depending on the actual experiment set-up. There is also a suite of advanced features to support your analysis, including:

- **Image-Based Auto-Trigger:** Trigger the camera (or even a number of connected cameras) from motion detected within the live image. This makes it easier to catch events that are not predictable and may occur infrequently.
- **Multi-Cine:** Partition internal memory into up to 63 segments and make shorter recordings back-to-back without missing any action.
- **Continuous Recording:** Perfect to record many occurrences of an event, especially an event that happens rarely or is unpredictable. Continuous recording mode automatically saves a cine to a connected PC immediately after it is recorded then re-arms the camera, waiting for the next cine.
  A recording can be triggered manually, from an event detection system, or even by our Image-Based Auto-Trigger. The number of recordings is limited only by the amount of available disk storage.

Trim Save the Cine in either its raw format, or convert it **popular formats such as Quicktime or AVI** or save frames in JPEG or TIFF, to easily email the analysis to colleagues.



### Additional Features:

Image-Based Auto-Trigger (IBAT)
Continuous Recording
Auto-Exposure
Multi-cine Acquisition
Gb Ethernet
Motion Analysis software included
Size and Weight:
  1.2 lb, 0.54 kg; 2.9 x 3.65 x 3.25 inches
  73 x 93 x 82.5 mm (H x W x D)
Operating Temperature: 0º C to 50º C
Tiered Service Contracts to protect your investment

**Maximum Frame Rates**

| Horizontal | Vertical | FPS |
|---|---|---|
| 1280 | 1024 | 915 |
| 1280 | 720 | 1295 |
| 768 | 768 | 1215 |
| 640 | 480 | 1935 |
| 512 | 512 | 1815 |
| 384 | 288 | 3180 |
| 256 | 256 | 3565 |
| 128 | 128 | 6870 |
| 128 | 8 | 52,445 |

### Focused

Since 1950, Vision Research has been designing, and manufacturing high-speed cameras. Our single focus is to invent, build, and support the most advanced cameras possible.

VISION RESEARCH

AMETEK®
MATERIALS ANALYSIS DIVISION

*100 Dey Road*
*Wayne, NJ 07470 USA*
*+1.973.696.4500*

www.phantomhighspeed.com